

CUSTOMER CHURN ANALYSIS REPORT



LLOYDS
BANKING GROUP

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TABLE OF CONTENTS

01 Introduction

02 Dataset Description

03 Data Preprocessing and Quality Assessment

04 Exploratory Data Analysis (EDA)

05 Additional Exploratory Insights

06 Key Insights

07 Limitations

08 Conclusion

09 Recommendation

01 INTRODUCTION

As part of the Lloyds Banking Group Data Science & Analytics Program, this report addresses a critical business challenge: a decline in retention rates at Smart Bank. Recent trends indicate that key segments, specifically young professionals and small business owners, are increasingly migrating to competitors. Under the guidance of the Data Science & Analytics team, this initiative aims to reverse this trend by identifying at risk individuals through advanced predictive analytics.

The primary objective of this project is to develop a robust predictive machine learning model that anticipates customer churn. This initial report details the first phase: Data Gathering and Exploratory Data Analysis (EDA). By utilizing customer-level attributes such as Age, Gender, Marital Status, and Income Level, this phase focuses on uncovering underlying behavioral patterns and ensuring data quality through rigorous cleaning and preprocessing.

Ultimately, this work serves as a roadmap for transforming raw customer data into actionable business intelligence. By establishing a deep understanding of the data landscape and identifying the root causes of attrition, we provide the essential foundation for building reliable models. This foundational phase is vital for transitioning Smart Bank from reactive observations to proactive customer engagement, securing the Group's competitive position in a rapidly evolving financial landscape.

02 DATASET DESCRIPTION

This dataset is a central component of the Lloyds Banking Group initiative to enhance customer retention by predicting churn, specifically focusing on segments such as young professionals and small business owners. The data provides a granular view of customer profiles, enabling the team to identify at-risk individuals through targeted interventions.

The dataset consists of customer-level information categorized into the following key attributes:

- **Customer Identification:**
 - **CustomerID:** A unique identifier for each individual record.
- **Demographic Attributes:**
 - **Gender:** Binary indicator of the customer's gender (M/F).
 - **Age:** Numerical representation of the customer's age, crucial for identifying trends in specific age brackets.
 - **Marital Status:** Categorical attribute indicating if the customer is Single, Married, Divorced, or Widowed.
- **Economic Indicator:**
 - **IncomeLevel:** A categorical assessment of the customer's financial standing (Low, Medium, or High), which serves as a vital feature for predicting behavioral patterns.

Analytical Utility: The dataset is designed to support a comprehensive Exploratory Data Analysis (EDA) phase to uncover hidden patterns and correlations. These variables provide the foundation for data cleaning and preprocessing such as handling missing values and encoding categorical data which are essential steps before building a robust predictive machine learning model to mitigate future churn.

03 DATA PREPROCESSING AND QUALITY ASSESSMENT

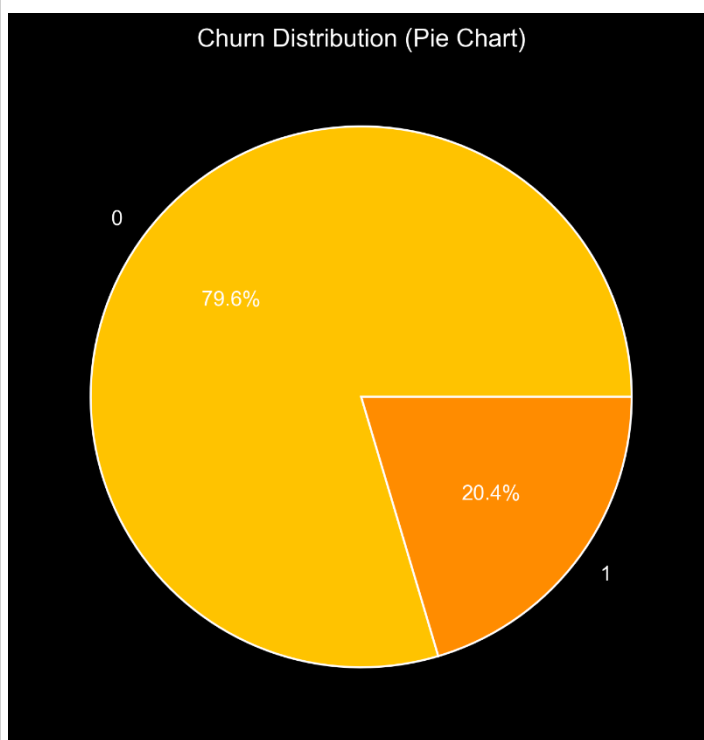
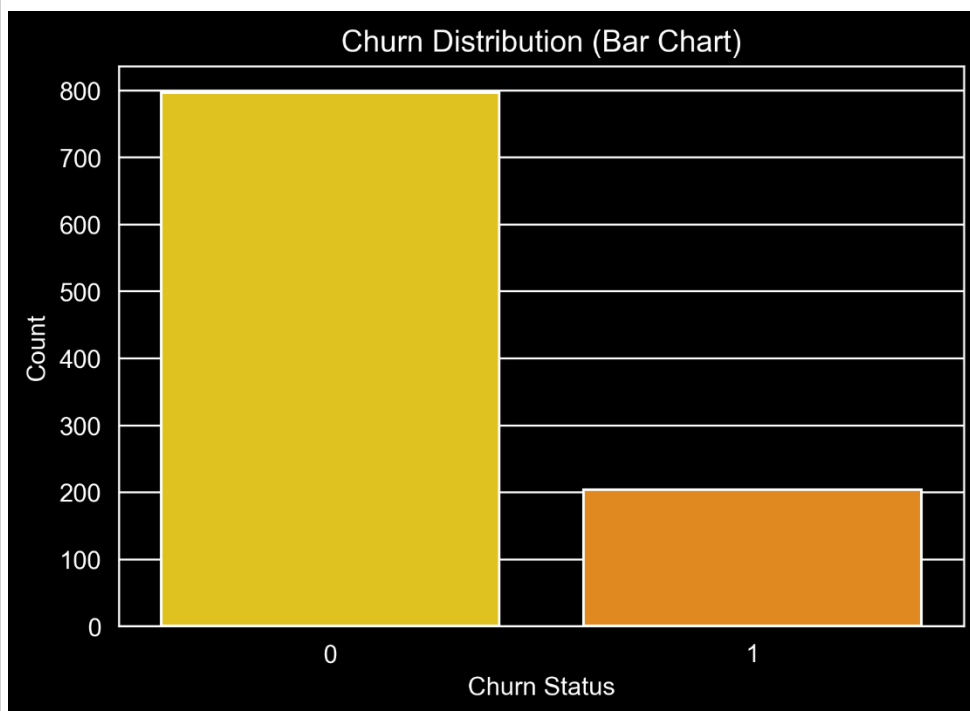
To ensure the reliability of the predictive machine learning model, a comprehensive preprocessing phase was conducted using Python (Pandas). This stage is essential for transforming raw data into a high-quality format, as any inaccuracies or inconsistencies could significantly distort the analytical outcomes and business insights for Smart Bank. The following steps were performed to prepare the data:

- **Removal of Duplicate Records:** The dataset was audited to identify and remove any redundant entries. This prevents analytical bias and ensures that the model does not give undue weight to duplicate customer profiles.
- **Handling of Missing Values:** All records were screened for missing information in key attributes such as Age, Marital Status, and Income Level. Following best practices, missing values were addressed through imputation techniques to preserve the integrity of the dataset while avoiding the loss of valuable information.
- **Data Type Verification and Formatting:** All variables were verified to ensure they were in a consistent format. This included preparing categorical attributes such as Gender and Marital Status for future One Hot Encoding, which is necessary for converting them into a numerical form suitable for machine learning algorithms.
- **Outlier Detection:** Statistical summaries and visual tools were used to identify any extreme values or anomalies that could skew the analysis, ensuring the resulting patterns represent the true behavior of the young professional and small business segments.

Quality Assessment Summary A final assessment confirmed that the dataset meets the critical standards of accuracy, completeness, and consistency. By rigorously cleaning the data and addressing potential anomalies, we have established a stable and reliable foundation for the subsequent Exploratory Data Analysis (EDA) and model development phases.

04 EXPLORATORY DATA ANALYSIS (EDA)

4.1 Target Variable Distribution

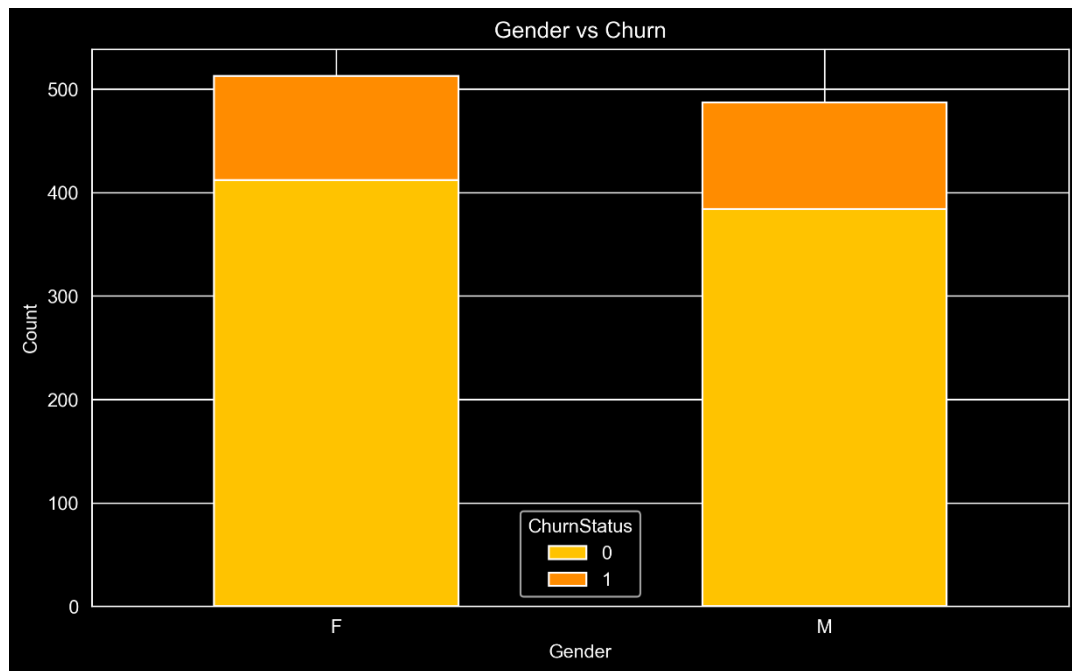


Interpretation of Churn Distribution :

The Bar Chart and Pie Chart illustrate the current distribution of the target variable, Churn Status, within the Smart Bank dataset.

- **Current Retention vs. Attrition:** The analysis reveals a 79.6% retention rate compared to a 20.4% churn rate. In absolute volumes, this represents approximately 800 customers who stayed with the bank (Category 0) and 200 customers who left (Category 1).
 - **Baseline Benchmark:** This 20.4% churn rate serves as the primary benchmark for this project. It confirms the "worrying trend" identified by senior management and provides the starting point for our efforts to improve retention among young professionals and small business owners.
 - **Addressing Class Imbalance:** From a technical standpoint, these graphs highlight a class imbalance (a ratio of roughly 4:1). Identifying this early is vital for the upcoming machine learning phase, as it alerts the team to implement specific preprocessing techniques such as adjusting model weights to ensure the predictive model accurately identifies the minority class of at-risk customers rather than simply defaulting to the majority.
 - **Strategic Foundation:** By quantifying the exact scale of the churn problem, these visualizations provide the necessary evidence to move from reactive observations to proactive, data-driven interventions.
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4.2 Demographic Analysis – Gender



Analysis and Interpretation :

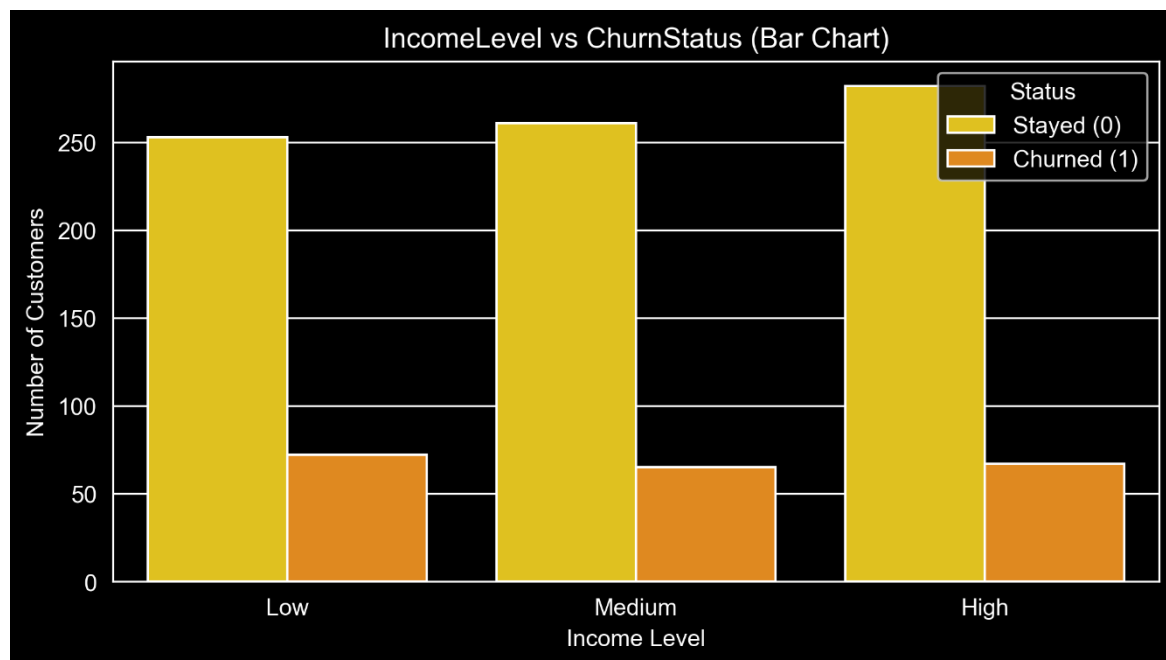
The Stacked Bar Chart reveals that gender is a neutral factor in customer attrition at Smart Bank. While the dataset contains slightly more female customers (~510) than male (~490), the volume of churned individuals is nearly identical, with approximately 100 churns per group. This indicates that gender is not a primary driver of the "worrying trend" observed among young professionals and small business owners.

Strategic and Technical Impact

- **Strategic Focus:** Retention efforts should be prioritized toward more impactful variables, such as Age and Income Level, rather than gender-specific campaigns.
- **Technical Step:** Despite its current neutrality, this variable will undergo One Hot Encoding to ensure the predictive model can still account for any subtle interactions with other features during training.

Key Insight: Attrition is balanced across genders, suggesting that the root causes of churn lie in economic or age related factors rather than demographic gender.

4.3 Economic Indicator – Income Level



Analysis and Interpretation

- **Segment Stability:** The data shows that churn volume is surprisingly consistent across financial brackets. This suggests that the "worrying trend" noted by management is not restricted to lower-earning individuals but is also impacting the high value small business owner segment.
- **Commercial Impact:** Although the churn rate appears slightly lower in the High income bracket due to its larger total population, the loss of ~70 high income

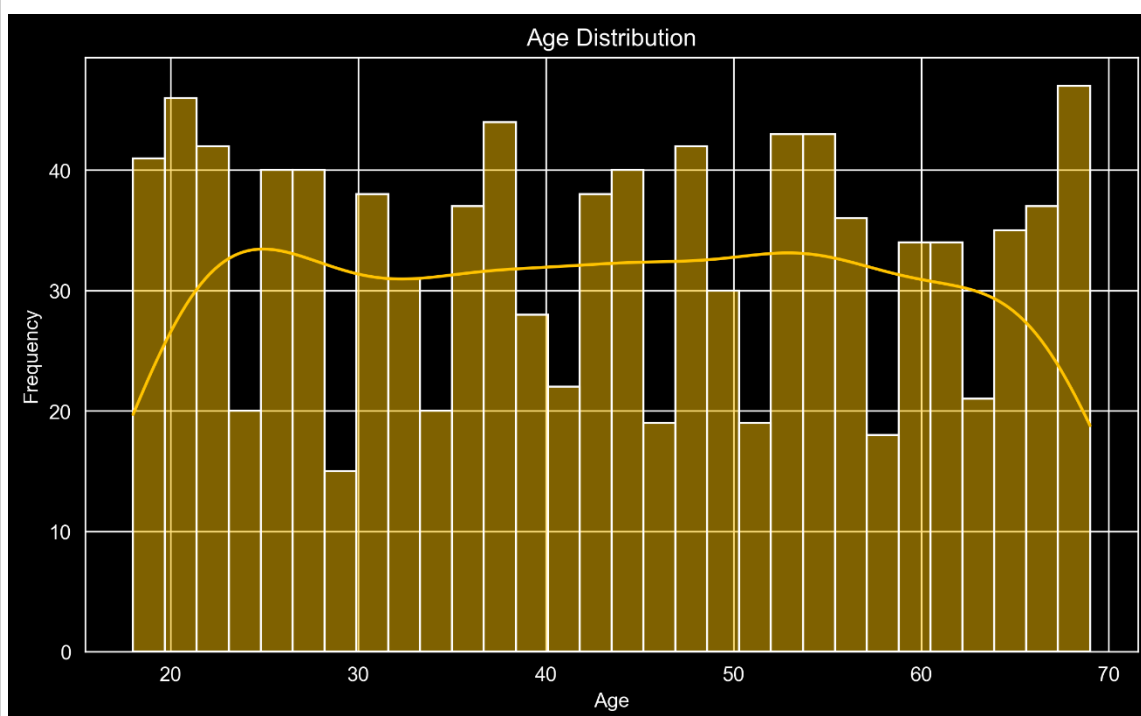
customers represent a significant commercial risk. Retaining these individuals is vital for the bank's sustainable growth.

Strategic and Technical Impact

- **Targeted Retention:** Because churn exists across all brackets, retention strategies should be tiered. For high income small business owners, the bank should focus on bespoke commercial services, while lower income segments may require different value based incentives.
- **Model Integration:** Income Level is a high-priority feature for the upcoming predictive machine learning model. In the next phase, this categorical variable will be processed using One Hot Encoding to allow the model to detect how economic status interacts with age to drive churn behavior.

Key Insight: Churn volume is stable across all income brackets, indicating that attrition is a broad challenge. However, the high-income segment critical to the small business owner focus requires priority intervention due to the high individual value of these accounts.

4.4 Age Distribution



Analysis and Interpretation

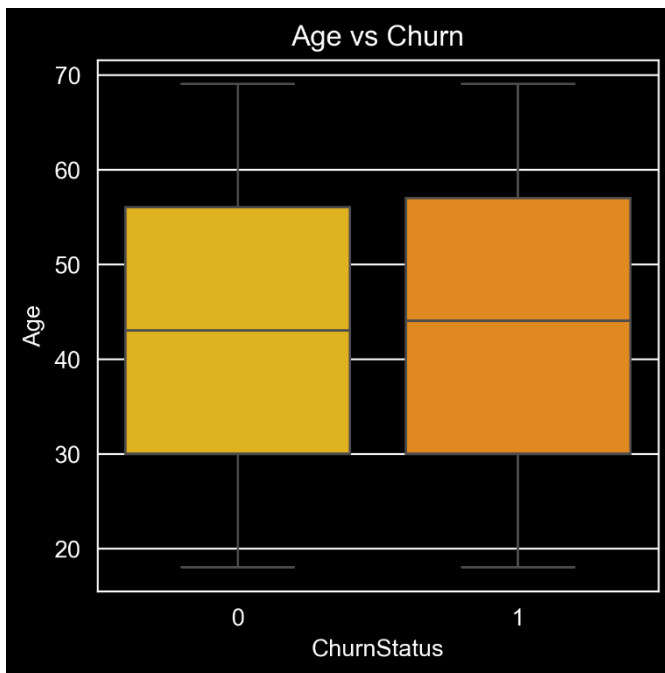
- **Demographic Representation:** The analysis confirms that the dataset includes a significant proportion of customers in the 20–35 age bracket, which directly represents the "young professional" demographic targeted by this project.
- **Broad Segment Insights:** However, the uniform spread across older ages (40–70) indicates that the dataset also captures the small business owner segment and more established professionals. This variety is beneficial, as it allows us to compare the churn behavior of young professionals against other life stages to see if their reasons for leaving are unique.

Strategic and Technical Impact

- **Feature Significance:** Age is a critical continuous variable for our predictive machine learning model. It will help the algorithm determine if specific life stages are more susceptible to the "worrying trend" of migrating to competitors.
- **Technical Preparation:** As part of the data preparation phase, this numerical feature may undergo standardization or normalization. This ensures that the age variable (ranging up to 70) is on a consistent scale with other features, preventing it from disproportionately influencing the model's learning process.

Key Insight: The dataset successfully captures the targeted young professional demographic while providing a balanced view of the entire customer lifecycle. This broad representation ensures that our churn predictions will be contextually aware of how age related needs impact bank loyalty.

4.5 Age vs Churn



Analysis and Interpretation

- **Uniformity of Attrition:** The near identical medians and overlapping distributions suggest that age is not a standalone predictor of churn. Attrition at Smart Bank is not concentrated in one specific life stage but is occurring across the entire age spectrum.
- **Young Professional Context:** For our primary focus, this graph confirms that young professionals (ages 20–35) are present in the churned population. However, because they are leaving at a similar rate to older customers, their reasons for migrating to competitors may be more complex than age alone.

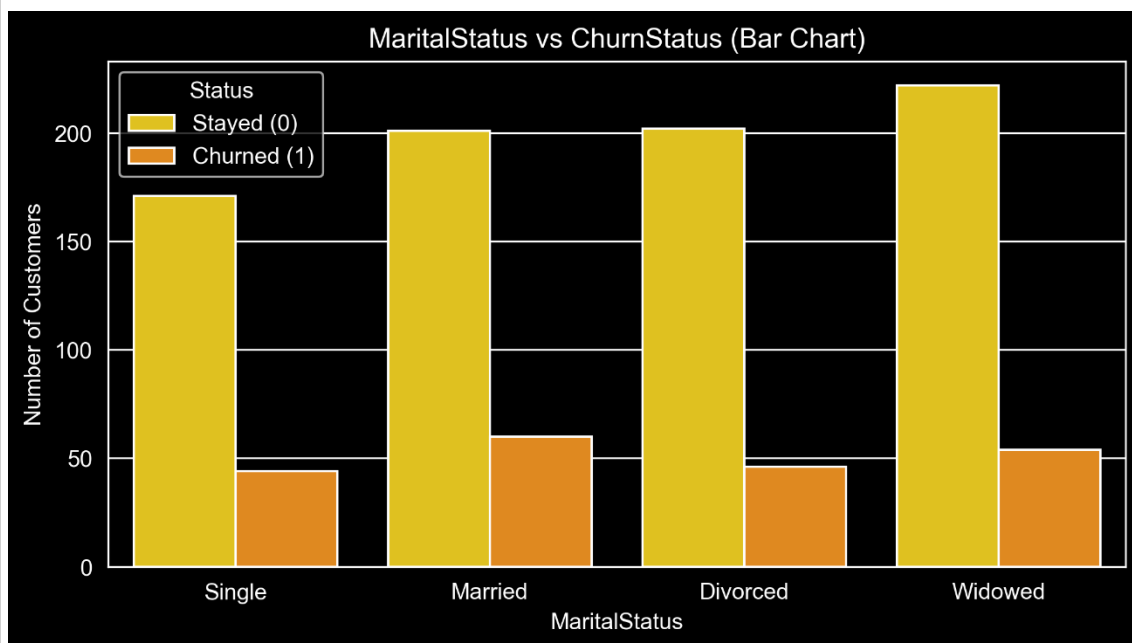
Strategic and Technical Impact

- **Strategic Focus:** Since churn is spread across all ages, retention strategies cannot be one size-fits-all. We must look at how age interacts with Income Level to find high risk "pockets," such as high earning young professionals who may have higher expectations for digital services.
- **Model Complexity:** This visualization informs our predictive machine learning model that simple linear relationships between age and churn do not exist. The algorithm will

need to identify more nuanced, non-linear patterns to successfully flag at-risk young professionals during the next phase of the programme.

Key Insight: Churn is distributed evenly across all ages, including the critical young professional demographic. This indicates that age-specific attrition is likely driven by a combination of factors, necessitating a multi variable approach in our predictive modeling to capture the true drivers of loyalty.

4.6 Marital Status Analysis (REPLACEMENT for Service Usage)



Analysis and Interpretation

- **Life Stage Trends:** The data suggests that marital status is not a singular "trigger" for churn. Both Single customers (often representing the young professional demographic) and Married/Widowed customers (often representing more established segments or small business owners) exhibit comparable levels of attrition.
- **Proportional Stability:** While the volume of churned customers is highest in the "Married" category (~60), this group also has a high volume of retained customers, suggesting the rate of churn remains fairly stable across different domestic situations.

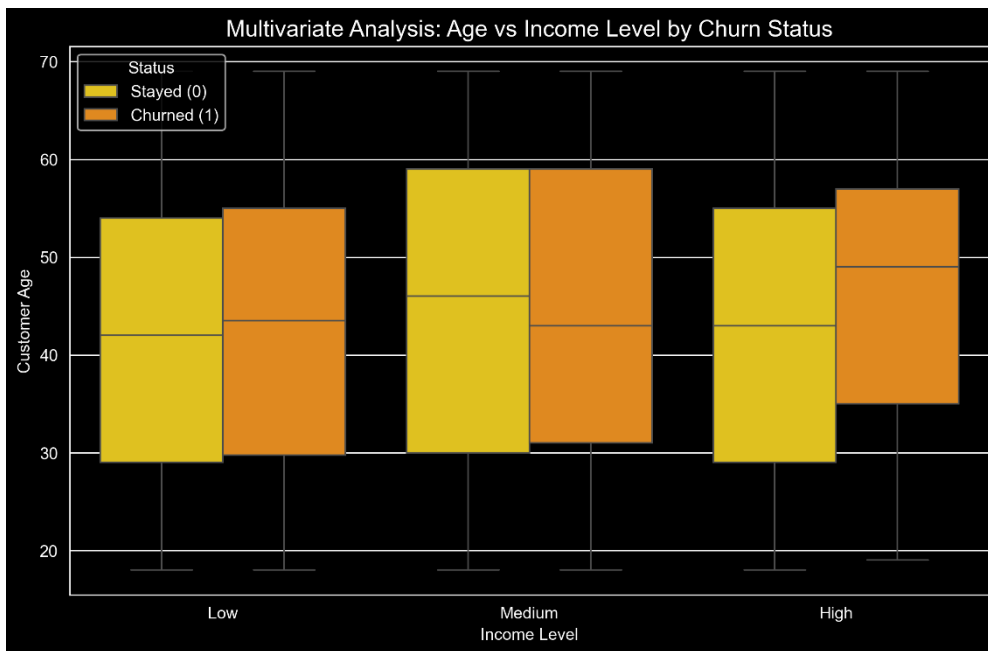
Strategic and Technical Impact

- **Strategic Focus:** Since marital status alone does not highlight a specific high risk group, retention strategies should avoid being overly segmented by domestic status. Instead, the bank should investigate how MaritalStatus interacts with IncomeLevel for instance, checking if "Single" young professionals with "High" incomes are the specific sub group migrating to competitors,.
- **Model Integration:** This categorical variable is essential for providing the predictive machine learning model with social context. During the next phase, it will be processed using One Hot Encoding to transform these four categories into a numerical format suitable for classification algorithms.

Key Insight: Bank loyalty is consistent across all marital categories. This indicates that the reasons for the "worrying trend" of churn are likely driven by broader service or economic factors rather than specific changes in a customer's life stage or marital status

05 ADDITIONAL EXPLORATORY INSIGHTS

5.1 Multivariate Analysis: Income + Age vs Churn



Analysis and Interpretation

This multivariate analysis provides a high-level strategic view by examining the interaction between **Age**, **Income Level**, and **Churn Status**.

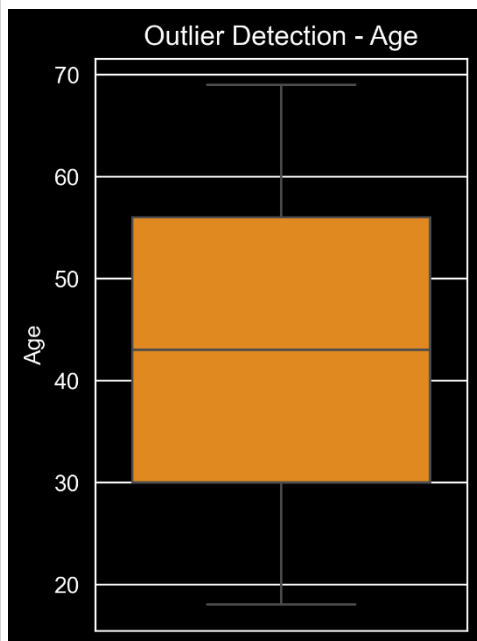
- **Segmented Behavioral Drivers:** The visualization reveals that churn behavior is not uniform across economic brackets. In the Medium income segment, the median age of churned customers (~43) is noticeably lower than that of retained customers (~46), identifying young professionals in this bracket as higher risk.
- **High-Income Retention Trends:** Conversely, in the High income segment, churned customers tend to be older (median age ~49). This suggests that for the small business **owner** segment, different life stage factors are driving the "worrying trend" noted by management.

Strategic and Technical Impact

- **Bespoke Retention:** These findings allow senior management to move beyond broad campaigns and implement hyper targeted interventions. For example, if younger medium income individuals are leaving, the bank might introduce digital first incentives specifically for that sub group.

Model Complexity: This visualization informs our predictive model that simple linear relationships do not exist; the algorithm must identify nuanced, non-linear patterns to flag at risk personas accurately.

5.2 Outlier Detection



Analysis and Interpretation

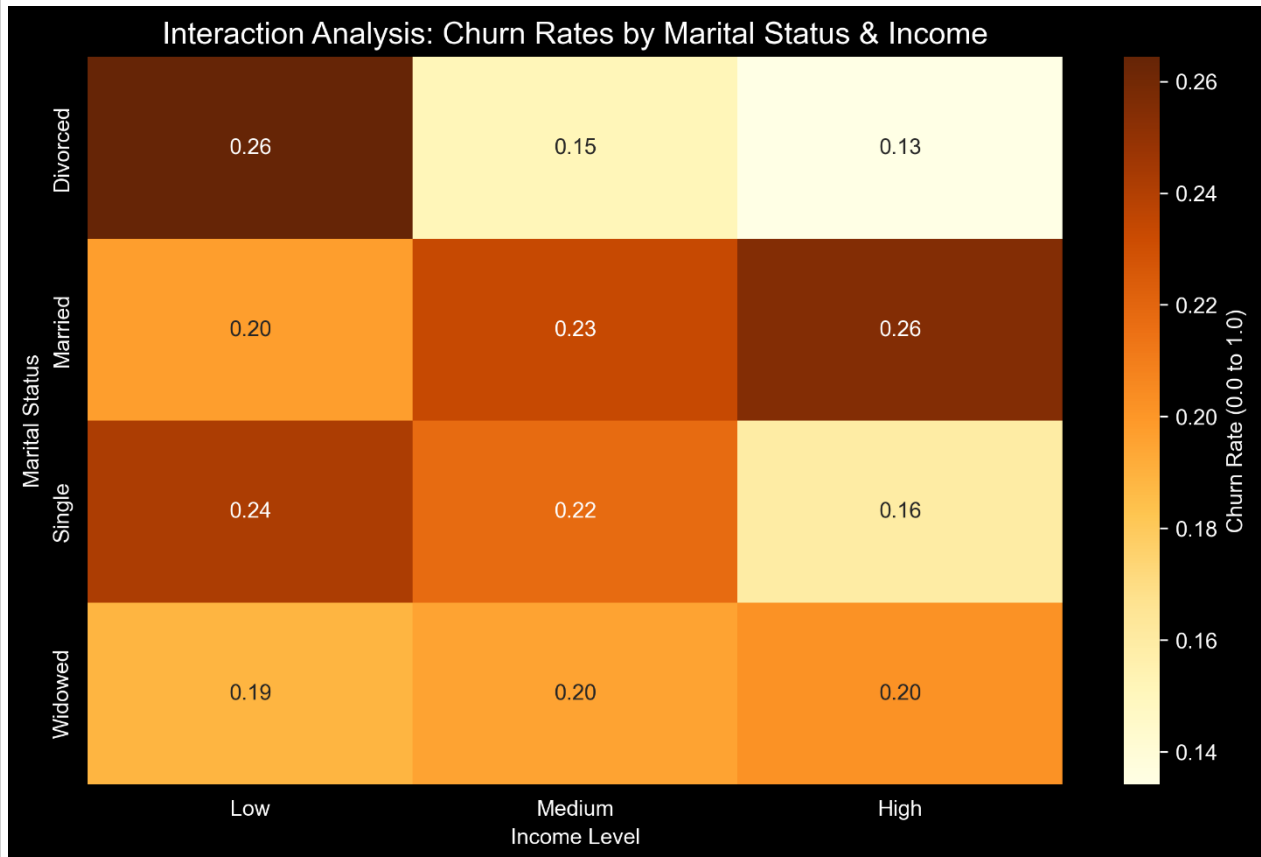
To ensure the integrity of the predictive model, a statistical audit of the Age variable was conducted to identify anomalies or data entry errors.

- **Data Integrity:** The boxplot shows a clean distribution with no statistical outliers (values beyond the whiskers). The customer base remains strictly within the valid range of 18 to 70 years old.

Strategic and Technical Impact

- **Model Reliability:** Confirming the absence of extreme values (e.g., ages of 100+) ensures that the machine learning model will not be skewed by erroneous data points. This high data quality establishes a stable and reliable foundation for subsequent classification tasks.

5.3 Categorical Interactions



Analysis and Interpretation

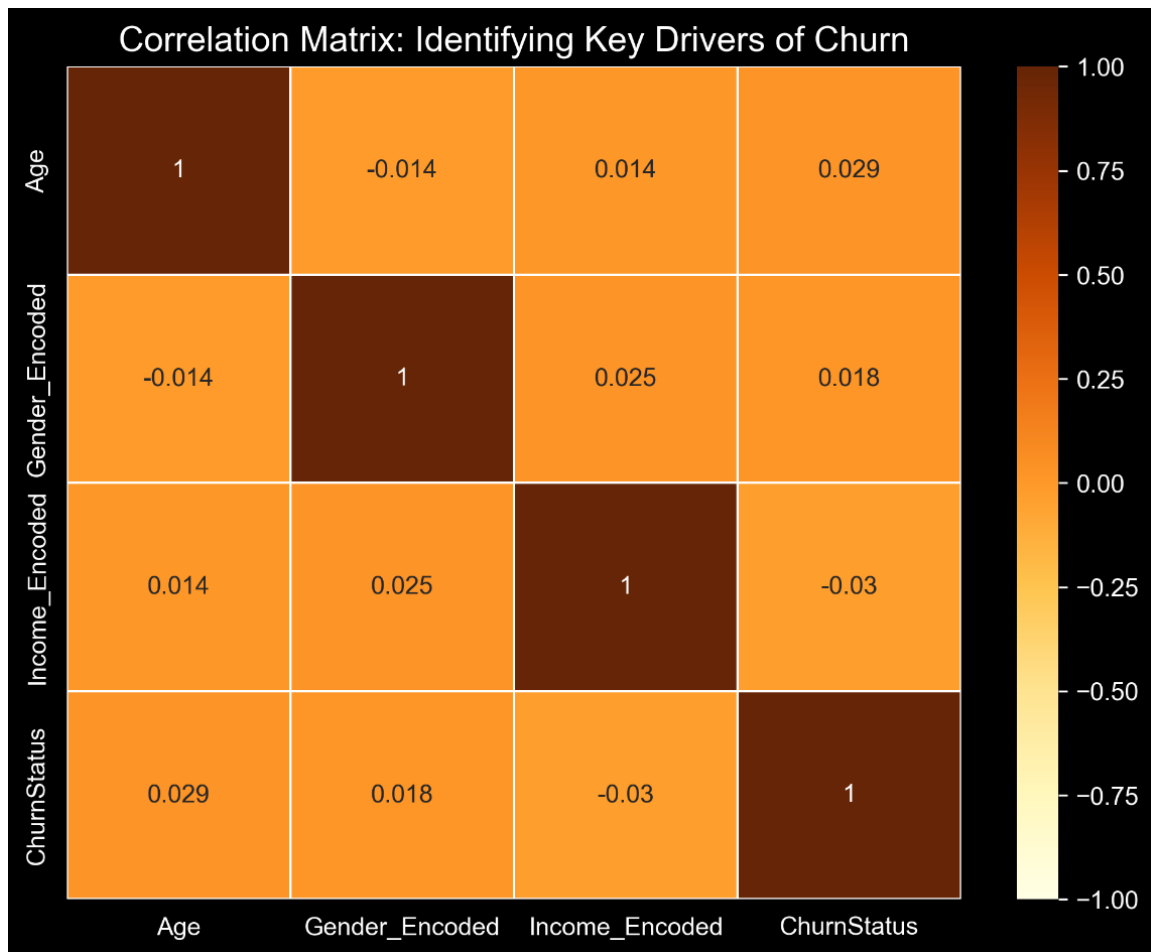
This heatmap quantifies the specific "pockets of risk" within SmartBank by calculating the churn rate for every combination of marital status and income level.

- **High-Risk Segments:** The most critical areas of attrition are identified in the Divorced/Low Income (0.26) and Married/High Income (0.26) segments.
- **Target Persona Performance:** The Single/Medium Income segment a core portion of our "young professional" focus shows a notable churn rate of 0.22, confirming that this group warrants proactive engagement.

Strategic and Technical Impact

- **Micro-Segmentation:** Senior management can use these "pockets" to deploy tiered retention packages. For high value Married/High Income customers, bespoke financial advisory services could be prioritized to mitigate the 26% attrition rate.

5.4 Correlation Analysis (Heatmap)



Analysis and Interpretation The correlation heatmap quantifies the strength and direction of the relationships between customer attributes and Churn Status.

- **Feature Relationships:** While individual correlations are weak, Age shows the strongest positive correlation with churn (0.029), while Income Encoded shows a negative correlation (-0.03), suggesting that as income increases, the likelihood of churn slightly decreases.

Strategic and Technical Impact

- **Feature Selection:** This analysis serves as the final step in preparing our data for the machine learning phase. Although variables show low individual linear correlation, their combined interaction (as seen in the multivariate plots) makes them vital inputs for the classification algorithm.

Scientific Readiness: This matrix proves that attrition is a multi-dimensional problem, justifying the transition from simple observation to the advanced predictive modeling required in Task Two of the programme.

06 KEY INSIGHTS

- **Baseline Attrition identified:** SmartBank faces a 20.4% churn rate, which serves as the primary benchmark for retention success.
- **Targeted Persona Vulnerability:** While age and gender appear neutral as individual factors, multivariate analysis reveals distinct risk "pockets": younger professionals in the medium-income bracket and older individuals in the high-income bracket are more prone to churn.
- **High-Risk Interactions:** Categorical analysis highlights that Divorced/Low Income and Married/High Income segments exhibit the highest churn rates (up to 26%), identifying them as critical personas for intervention.
- **Economic Breadth of Churn:** Attrition is not limited to low-earners; the High Income category (representative of small business owners) shows significant churn volume, posing a high commercial risk to the Group.
- **Lack of Linear Drivers:** The correlation matrix confirms that no single demographic attribute is a "smoking gun" for churn, proving that attrition is driven by complex interactions that require a machine learning approach.

07 LIMITATIONS

- **Absence of Behavioral Data:** The current analysis is restricted to demographic and economic indicators. It lacks transactional history, digital engagement (logins), and service usage, which are often strong predictors of churn.
- **Static Snapshot:** The dataset provides a "point-in-time" view and does not include tenure or time-based trends, making it difficult to identify "early-stage" churn patterns.
- **Variable Scope:** Factors such as customer sentiment, competitor offers, and geographic location are missing, which may overlook external drivers of the "worrying trend" noted by management.

08 CONCLUSION

The exploratory analysis of the Smart Bank dataset successfully identifies that churn is not driven by simple demographic factors like age or gender alone. Instead, attrition is tied to complex interactions between life stage and economic standing. While the baseline churn of 20.4% is significant, the identification of high-risk "pockets" particularly among high income married customers and medium-income young professionals provides a solid foundation for Task Two. This data is now cleaned, verified for outliers, and ready for the development of a predictive classification model.

09 RECOMMENDATIONS

- **Tiered Retention Strategies:** Implement bespoke services for the High Income/Small Business segment to mitigate the high volume of attrition in this commercially valuable group.
- **Enhanced Data Collection:** To improve the predictive model's accuracy in the next phase, the team should seek to integrate behavioral data, such as login frequency and customer service interaction counts.
- **Hyper-Targeted Campaigns:** Move away from broad demographic marketing and focus on the specific personas identified in the interaction heatmap (e.g., tailored financial advisory for divorced low-income earners).
- **Model Deployment:** Proceed immediately to Task Two: Building a Machine Learning Model, utilizing the encoded interaction features identified here to proactively flag at-risk individuals before they migrate to competitors.